APPLICATION OF QUANTUM COMPUTING IN DIGITAL EDUCATION SYSTEMS

Digital education systems generate a huge amount of student data, making personalized learning, performance prediction, and educational resource recommendation difficult to manage efficiently using traditional computing methods. We chose Quantum Computing (QC) because quantum techniques can explore complex solution spaces and are particularly suitable for optimization and high-dimensional educational problems. Existing strategies mainly use rule-based systems, classical machine learning, greedy algorithms, genetic algorithms, collaborative filtering, and integer programming, but these methods can become computationally expensive for large and complex problems. Our proposed solution is a hybrid classical-quantum framework, where classical computers manage data and regular operations while quantum algorithms solve selected complex optimization tasks. In particular, QAOA can be used to optimize personalized learning paths by representing the problem as a QUBO model. The quantum processor generates possible solutions, while classical processing checks prerequisites, constraints, and learning outcomes to select the most suitable path. The expected output is a personalized and optimized learning path for each student, along with improved educational recommendations and better support for student-performance analysis. The study indicates that quantum approaches may improve learning-path solution quality by approximately 15–25% for sufficiently complex problems, although the actual benefit depends on problem size and current quantum hardware capabilities.

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